Model Architecture: Llama-3.2-1B

Generated by sg_modelload

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Contents

1 Model Overview **2**

2 Mathematical Representation **2**
- 2.1 The model as a composition of layers . . . 2
- 2.2 The layers, top to bottom . . . 3
- 2.3 Neurons and attention heads . . . 4
- 2.4 Architecture and parameters . . . 4
- 2.5 Output representations . . . 4
- 2.6 Intermediate states and sub-network decomposition . . . 5
- 2.7 Local linear representation . . . 5
- 2.8 Summary of notation . . . 5

3 Forward Pass in Pseudocode **6**

4 Compact Architecture (repeated layers written once) **6**
- 4.1 Modules outside the repeated stacks . . . 6
- 4.2 Language-model decoder: `model.layers` . . . 6

5 What Each Layer Does **7**

6 Parameter Count **7**

A Full Layer-by-Layer Table **8**

B Per-Layer Parameter Count **11**

1 Model Overview

Model name	Llama-3.2-1B
Transformers class	`LlamaForCausalLM`
Path	`/models/memorization/Llama-3.2-1B`
Same architecture as its path	Llama-3.2-1B-Instruct `/models/sayak_da_work/Llama-3.2-1B-Instruct`
Total parameters	1,235,814,400 (1.24B)
Weight dtype	bfloat16
Hidden size d	2,048
Decoder layers L	16
Query heads H	32
Key/value heads H_{kv}	8
Head dim d_h	64
MLP width d_{ff}	8,192
Vocabulary V	128,256
Max context length	131,072
RMSNorm ϵ	1e-05
Tied input/output embeddings	True
Activation	SiLU
RoPE base θ	500,000

Same structure. Llama-3.2-1B-Instruct has exactly the same architecture as Llama-3.2-1B (same class, every tensor name and shape identical, checked when this document was generated). Fine-tuning changes only the weight values, so everything in this document applies to it unchanged. The two configs differ only in generation settings such as `eos_token_id`.

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (2048)
V	vocabulary size (128,256)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (Llama-3.2-1B as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$

($V = 128{,}256$). It is a composition of an embedding E, $L = 16$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E \tag{1}$$

Every block maps $\mathbb{R}^{T\times d} \to \mathbb{R}^{T\times d}$ ($d = 2048$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.layers.`$(\ell - 1)$.

Definition 2.2 (Block types)**.** All L blocks have the same functional form $\mathcal{B}_\mathrm{D}$ (Decoder block). They differ only in their weights: $\mathcal{B}^{(\ell)} = \mathcal{B}_\mathrm{D}(\,\cdot\,;\theta_\ell)$. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}_\mathrm{D}^{(16)} \circ \cdots \circ \mathcal{B}_\mathrm{D}^{(3)} \circ \mathcal{B}_\mathrm{D}^{(2)} \circ \mathcal{B}_\mathrm{D}^{(1)} \circ E \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \left(\mathcal{B}_\mathrm{D}^{\circ 16}\right) \circ E \tag{3}$$

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_\mathrm{D}$	Decoder block	16	60,821,504	0–15

Language-model decoder (16 layers). layers 0–15: $[\mathcal{B}_\mathrm{D}]$ repeated 16× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\mathrm{FFN}(h)_t = \mathrm{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_\mathrm{out}\times d_\mathrm{in}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.3 (Token embedding)**.** $E : \{1,\dots,V\}^T \to \mathbb{R}^{T\times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V\times d}$ per token:

$$E(x)_t = W_E[x_t, :], \qquad t = 1,\dots,T. \tag{4}$$

Definition 2.4 (Normalisation)**.** $\mathrm{RMSNorm}(u) = \gamma \odot u/\sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. $\mathrm{RMSNorm}_1$, $\mathrm{RMSNorm}_\mathrm{final}$).

Definition 2.5 (Block $\mathcal{B}_\mathrm{D}$: Decoder block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 2048}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_\mathrm{D} = (\mathrm{id} + \mathrm{FFN} \circ \mathrm{RMSNorm}_2) \circ (\mathrm{id} + \mathrm{Attn} \circ \mathrm{RMSNorm}_1) \tag{5}$$

Step by step, the sub-layers are:

$$\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\, u && H = 32 \text{ heads of size } d_h = 64\\
K,\ V &= W_K\, u, \quad W_V\, u && H_{kv} = 8 \text{ key/value heads}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\, Q_h K_{g(h)}^\top + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\, V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\,[O_1;\dots;O_H]\\
\mathrm{FFN}(u) &= W_\mathrm{down}\big(\,\mathrm{SiLU}(W_\mathrm{gate}\,u) \odot W_\mathrm{up}\,u\big) && \text{gated MLP, width 8192}
\end{aligned}$$

Parameters of one block ($|\theta_\ell| = 60{,}821{,}504$): $W_Q \in \mathbb{R}^{2048\times 2048}$, $W_K \in \mathbb{R}^{512\times 2048}$, $W_V \in \mathbb{R}^{512\times 2048}$, $W_O \in \mathbb{R}^{2048\times 2048}$, $W_\mathrm{gate} \in \mathbb{R}^{8192\times 2048}$, $W_\mathrm{up} \in \mathbb{R}^{8192\times 2048}$, $W_\mathrm{down} \in \mathbb{R}^{2048\times 8192}$, plus 4,096 parameters in norm scales, biases and other vectors. This type is used 16 times.

Definition 2.6 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token, re-using the embedding matrix (tied weights):

$$\Psi(h)_t = W_E^\top\ \mathrm{RMSNorm}_\mathrm{final}(h)_t \tag{6}$$

2.3 Neurons and attention heads

Definition 2.7 (Neuron, pre-activation value and state)**.** The feed-forward sub-layer of block ℓ contains $d_{ff}^{(\ell)} = 8192$ neurons. Neuron j of layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_j^{(\ell)}(u) = \mathrm{SiLU}\left(W_{\mathrm{gate},j}^{(\ell)} \cdot u\right)\left(W_{\mathrm{up},j}^{(\ell)} \cdot u\right), \tag{7}$$

where $W_{\mathrm{gate},j}$ is the j-th row of W_{gate}. The feed-forward output is the sum of the neuron outputs, each written along its own column of W_{down}:

$$\mathrm{FFN}_\ell(u) = \sum_{j=1}^{d_{ff}^{(\ell)}} \eta_j^{(\ell)}(u)\, W_{\mathrm{down}}^{(\ell)}[:, j]. \tag{8}$$

Across its L blocks the network has $N = \sum_{\ell=1}^{L} d_{ff}^{(\ell)} = 131{,}072$ feed-forward neurons. For input x at position t, with $u_t^{(\ell)}(x)$ the normalised input of the feed-forward sub-layer:

- $\mathcal{V}(\eta; x, t) = W_{\mathrm{gate},j}^{(\ell)} \cdot u_t^{(\ell)}(x) \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \le 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.8 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T \times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\mathrm{Attn}_\ell(u) = \sum_{h=1}^{H} P_h^{(\ell)}\, V_{g(h)}\, W_{O,h}^{(\ell)\top}. \tag{9}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 16 attention layers $\times$ $H = 32$ heads $= 512$ attention heads.

2.4 Architecture and parameters

Definition 2.9 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}) = (128{,}256,\ 2048,\ 16,\ (\mathrm{D}^{16}),\ 32,\ 8,\ 64,\ 8192). \tag{10}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\mathrm{final}}\}$ (W_E is shared with the output head). Grouping the blocks by type,

$$\begin{aligned} |\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\mathrm{rest}}| \\ &= 16 \times 60{,}821{,}504 + 262{,}670{,}336 \\ &= 1{,}235{,}814{,}400, \end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, ...; Table 3).

2.5 Output representations

Definition 2.10 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\left(x_{t+1} = k \mid x_1, \ldots, x_t\right)$.
3. **Predicted next token:** $z(x) = \mathrm{argmax}_{k \in \{1,\ldots,V\}}\, y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \ldots, x_t$, because every attention mask hides later positions. Training minimises the cross-entropy $-\sum_t \log p(x)_{t, x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \dots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E}_{F_{\ell-1}\ \text{(prefix)}} \tag{11}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E(x)$.
- **Inside layer ℓ:** the attention write $a^{(\ell)} = \mathrm{Attn}_\ell\big(\mathrm{RMSNorm}_1(h^{(\ell-1)})\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \mathrm{RMSNorm}_2(\hat{h}^{(\ell)})$, the pre-activations $\hat{x}^{(\ell)} = u^{(\ell)} W_{\text{gate}}^{(\ell)\top} \in \mathbb{R}^{T \times d_{ff}}$ and the neuron outputs $\eta^{(\ell)}(u^{(\ell)}) \in \mathbb{R}^{T \times d_{ff}}$, and the feed-forward write $m^{(\ell)} = \mathrm{FFN}_\ell(u^{(\ell)})$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = h^{(0)} + \sum_{\ell=1}^{L} \big(a^{(\ell)} + m^{(\ell)}\big) \tag{12}$$

So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, SiLU and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.11 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\mathrm{id} + g_{\ell,i})$ gives

$$J_\ell = (I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{13}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{14}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T \times V}$
$E,\ \Psi$	embedding and output head
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \dots, L$
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\mathrm{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ ($T \times T$)

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Symbol	Meaning
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell-1$) and suffix (layer $\ell+1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 Llama-3.2-1B: full forward pass

Require: token ids x of shape $B \times T$

$h \leftarrow E(x)$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}2048$

for $\ell \leftarrow 1$ **to** 16 **do** ▷ 16 blocks of type $\mathcal{B}_{\mathrm{D}}$

  $h \leftarrow \mathcal{B}_{\mathrm{D}}(h;\, \theta_\ell)$ ▷ Decoder block; result is $h^{(\ell)}$

end for

$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ

$y \leftarrow h\, W_E^\top$ ▷ logits $B{\times}T{\times}128,256$, embedding matrix re-used

return y ▷ softmax(y) = next-token probabilities

Algorithm 2 Block $\mathcal{B}_{\mathrm{D}}(h)$ — Decoder block

Require: h of shape $B \times T \times 2048$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection

$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector

token mixing: multi-head self-attention

$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}32{\times}64$ (32 query heads)

$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}8{\times}64$ (8 key/value heads, each shared by 4 query heads)

$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs

$P \leftarrow \mathrm{softmax}(s\, QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens

$O \leftarrow PV$ ▷ $B{\times}T{\times}32{\times}64$

$m \leftarrow O\, W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}2048$

$h \leftarrow r + m$ ▷ add residual

channel mixing: feed-forward network (each token independently)

$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$

$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big) W_{\mathrm{down}}^\top$ ▷ expand to 8192, gate, shrink back to 2048

$h \leftarrow r + m$ ▷ add residual

return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.embed_tokens.weight`	Embedding	128256 × 2048	262,668,288
`model.norm.weight`	LlamaRMSNorm	2048	2,048
`lm_head.weight` (tied to `model.embed_tokens.weight`)	Linear	128256 × 2048	0 (shared)

4.2 Language-model decoder: `model.layers`

Table 4: $\mathcal{B}_{\mathrm{D}}$: Decoder block — layers 0–15; weight names are relative to `model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	2048×2048	4,194,304
`self_attn.k_proj.weight`	Linear	512×2048	1,048,576
`self_attn.v_proj.weight`	Linear	512×2048	1,048,576
`self_attn.o_proj.weight`	Linear	2048×2048	4,194,304
`mlp.gate_proj.weight`	Linear	8192×2048	16,777,216
`mlp.up_proj.weight`	Linear	8192×2048	16,777,216
`mlp.down_proj.weight`	Linear	2048×8192	16,777,216
`input_layernorm.weight`	LlamaRMSNorm	2048	2,048
`post_attention_layernorm.weight`	LlamaRMSNorm	2048	2,048
Per block			**60,821,504**
$\times$ 16 blocks			**973,144,064**

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
Language model	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?".
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.

6 Parameter Count

Table 6: Parameters by component

Component	Parameters	% of total
MLP / MoE	805,306,368	65.2%
Token embedding	262,668,288	21.3%

continued on next page

Component	Parameters	% of total
Attention / token mixer	167,772,160	13.6%
Norms	67,584	0.0%
Total	**1,235,814,400**	**100.0%**

Table 7: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{D}}$ Decoder block	16	60,821,504	973,144,064
Modules outside stacks	1	–	262,670,336
Total			**1,235,814,400**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	128256 × 2048	bfloat16	262,668,288
`model`	LlamaModel	–	–	–
`model.embed_tokens.weight`	Embedding	128256 × 2048	bfloat16	262,668,288
`model.layers.0`	LlamaDecoderLayer	–	–	–
`model.layers.0.input_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.0.mlp`	LlamaMLP	–	–	–
`model.layers.0.mlp.down_proj.weight`	Linear	2048 × 8192	bfloat16	16,777,216
`model.layers.0.mlp.gate_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.0.mlp.up_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.0.post_attention_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.0.self_attn`	LlamaAttention	–	–	–
`model.layers.0.self_attn.k_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.0.self_attn.o_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.0.self_attn.q_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.0.self_attn.v_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.1`	LlamaDecoderLayer	–	–	–
`model.layers.1.input_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.1.mlp`	LlamaMLP	–	–	–
`model.layers.1.mlp.down_proj.weight`	Linear	2048 × 8192	bfloat16	16,777,216
`model.layers.1.mlp.gate_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.1.mlp.up_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.1.post_attention_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.1.self_attn`	LlamaAttention	–	–	–
`model.layers.1.self_attn.k_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.1.self_attn.o_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.1.self_attn.q_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.1.self_attn.v_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.2`	LlamaDecoderLayer	–	–	–
`model.layers.2.input_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.2.mlp`	LlamaMLP	–	–	–
`model.layers.2.mlp.down_proj.weight`	Linear	2048 × 8192	bfloat16	16,777,216
`model.layers.2.mlp.gate_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.2.mlp.up_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.2.post_attention_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.2.self_attn`	LlamaAttention	–	–	–
`model.layers.2.self_attn.k_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.2.self_attn.o_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.2.self_attn.q_proj.weight`	Linear	2048 × 2048	bfloat16	4,194,304
`model.layers.2.self_attn.v_proj.weight`	Linear	512 × 2048	bfloat16	1,048,576
`model.layers.3`	LlamaDecoderLayer	–	–	–
`model.layers.3.input_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.3.mlp`	LlamaMLP	–	–	–
`model.layers.3.mlp.down_proj.weight`	Linear	2048 × 8192	bfloat16	16,777,216
`model.layers.3.mlp.gate_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.3.mlp.up_proj.weight`	Linear	8192 × 2048	bfloat16	16,777,216
`model.layers.3.post_attention_layernorm.weight`	LlamaRMSNorm	2048	bfloat16	2,048
`model.layers.3.self_attn`	LlamaAttention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.3.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.3.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.3.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.3.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.4	LlamaDecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.4.mlp	LlamaMLP	–	–	–
model.layers.4.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.4.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.4.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.4.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.4.self_attn	LlamaAttention	–	–	–
model.layers.4.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.4.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.4.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.4.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.5	LlamaDecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.5.mlp	LlamaMLP	–	–	–
model.layers.5.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.5.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.5.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.5.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.5.self_attn	LlamaAttention	–	–	–
model.layers.5.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.5.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.5.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.5.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.6	LlamaDecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.6.mlp	LlamaMLP	–	–	–
model.layers.6.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.6.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.6.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.6.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.6.self_attn	LlamaAttention	–	–	–
model.layers.6.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.6.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.6.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.6.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.7	LlamaDecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.7.mlp	LlamaMLP	–	–	–
model.layers.7.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.7.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.7.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.7.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.7.self_attn	LlamaAttention	–	–	–
model.layers.7.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.7.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.7.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.7.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.8	LlamaDecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.8.mlp	LlamaMLP	–	–	–
model.layers.8.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.8.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.8.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.8.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.8.self_attn	LlamaAttention	–	–	–
model.layers.8.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.8.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.8.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.8.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.9	LlamaDecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.9.mlp	LlamaMLP	–	–	–
model.layers.9.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.9.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216

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Layer path	Type	Shape	Dtype	Params
model.layers.9.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.9.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.9.self_attn	LlamaAttention	–	–	–
model.layers.9.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.9.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.9.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.9.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.10	LlamaDecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.10.mlp	LlamaMLP	–	–	–
model.layers.10.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.10.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.10.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.10.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.10.self_attn	LlamaAttention	–	–	–
model.layers.10.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.10.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.10.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.10.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.11	LlamaDecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.11.mlp	LlamaMLP	–	–	–
model.layers.11.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.11.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.11.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.11.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.11.self_attn	LlamaAttention	–	–	–
model.layers.11.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.11.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.11.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.11.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.12	LlamaDecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.12.mlp	LlamaMLP	–	–	–
model.layers.12.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.12.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.12.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.12.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.12.self_attn	LlamaAttention	–	–	–
model.layers.12.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.12.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.12.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.12.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.13	LlamaDecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.13.mlp	LlamaMLP	–	–	–
model.layers.13.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.13.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.13.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.13.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.13.self_attn	LlamaAttention	–	–	–
model.layers.13.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.13.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.13.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.13.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.14	LlamaDecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.14.mlp	LlamaMLP	–	–	–
model.layers.14.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.14.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.14.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.14.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.14.self_attn	LlamaAttention	–	–	–
model.layers.14.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.14.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.14.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.14.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.15	LlamaDecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.layers.15.input_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.15.mlp	LlamaMLP	–	–	–
model.layers.15.mlp.down_proj.weight	Linear	2048 × 8192	bfloat16	16,777,216
model.layers.15.mlp.gate_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.15.mlp.up_proj.weight	Linear	8192 × 2048	bfloat16	16,777,216
model.layers.15.post_attention_layernorm.weight	LlamaRMSNorm	2048	bfloat16	2,048
model.layers.15.self_attn	LlamaAttention	–	–	–
model.layers.15.self_attn.k_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.layers.15.self_attn.o_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.15.self_attn.q_proj.weight	Linear	2048 × 2048	bfloat16	4,194,304
model.layers.15.self_attn.v_proj.weight	Linear	512 × 2048	bfloat16	1,048,576
model.norm.weight	LlamaRMSNorm	2048	bfloat16	2,048

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.embed_tokens	262,668,288	262,668,288
model.layers.0.input_layernorm	2,048	262,670,336
model.layers.0.mlp.down_proj	16,777,216	279,447,552
model.layers.0.mlp.gate_proj	16,777,216	296,224,768
model.layers.0.mlp.up_proj	16,777,216	313,001,984
model.layers.0.post_attention_layernorm	2,048	313,004,032
model.layers.0.self_attn.k_proj	1,048,576	314,052,608
model.layers.0.self_attn.o_proj	4,194,304	318,246,912
model.layers.0.self_attn.q_proj	4,194,304	322,441,216
model.layers.0.self_attn.v_proj	1,048,576	323,489,792
model.layers.1.input_layernorm	2,048	323,491,840
model.layers.1.mlp.down_proj	16,777,216	340,269,056
model.layers.1.mlp.gate_proj	16,777,216	357,046,272
model.layers.1.mlp.up_proj	16,777,216	373,823,488
model.layers.1.post_attention_layernorm	2,048	373,825,536
model.layers.1.self_attn.k_proj	1,048,576	374,874,112
model.layers.1.self_attn.o_proj	4,194,304	379,068,416
model.layers.1.self_attn.q_proj	4,194,304	383,262,720
model.layers.1.self_attn.v_proj	1,048,576	384,311,296
model.layers.2.input_layernorm	2,048	384,313,344
model.layers.2.mlp.down_proj	16,777,216	401,090,560
model.layers.2.mlp.gate_proj	16,777,216	417,867,776
model.layers.2.mlp.up_proj	16,777,216	434,644,992
model.layers.2.post_attention_layernorm	2,048	434,647,040
model.layers.2.self_attn.k_proj	1,048,576	435,695,616
model.layers.2.self_attn.o_proj	4,194,304	439,889,920
model.layers.2.self_attn.q_proj	4,194,304	444,084,224
model.layers.2.self_attn.v_proj	1,048,576	445,132,800
model.layers.3.input_layernorm	2,048	445,134,848
model.layers.3.mlp.down_proj	16,777,216	461,912,064
model.layers.3.mlp.gate_proj	16,777,216	478,689,280
model.layers.3.mlp.up_proj	16,777,216	495,466,496
model.layers.3.post_attention_layernorm	2,048	495,468,544
model.layers.3.self_attn.k_proj	1,048,576	496,517,120
model.layers.3.self_attn.o_proj	4,194,304	500,711,424
model.layers.3.self_attn.q_proj	4,194,304	504,905,728
model.layers.3.self_attn.v_proj	1,048,576	505,954,304
model.layers.4.input_layernorm	2,048	505,956,352
model.layers.4.mlp.down_proj	16,777,216	522,733,568
model.layers.4.mlp.gate_proj	16,777,216	539,510,784
model.layers.4.mlp.up_proj	16,777,216	556,288,000
model.layers.4.post_attention_layernorm	2,048	556,290,048

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Module	Params	Running total
model.layers.4.self_attn.k_proj	1,048,576	557,338,624
model.layers.4.self_attn.o_proj	4,194,304	561,532,928
model.layers.4.self_attn.q_proj	4,194,304	565,727,232
model.layers.4.self_attn.v_proj	1,048,576	566,775,808
model.layers.5.input_layernorm	2,048	566,777,856
model.layers.5.mlp.down_proj	16,777,216	583,555,072
model.layers.5.mlp.gate_proj	16,777,216	600,332,288
model.layers.5.mlp.up_proj	16,777,216	617,109,504
model.layers.5.post_attention_layernorm	2,048	617,111,552
model.layers.5.self_attn.k_proj	1,048,576	618,160,128
model.layers.5.self_attn.o_proj	4,194,304	622,354,432
model.layers.5.self_attn.q_proj	4,194,304	626,548,736
model.layers.5.self_attn.v_proj	1,048,576	627,597,312
model.layers.6.input_layernorm	2,048	627,599,360
model.layers.6.mlp.down_proj	16,777,216	644,376,576
model.layers.6.mlp.gate_proj	16,777,216	661,153,792
model.layers.6.mlp.up_proj	16,777,216	677,931,008
model.layers.6.post_attention_layernorm	2,048	677,933,056
model.layers.6.self_attn.k_proj	1,048,576	678,981,632
model.layers.6.self_attn.o_proj	4,194,304	683,175,936
model.layers.6.self_attn.q_proj	4,194,304	687,370,240
model.layers.6.self_attn.v_proj	1,048,576	688,418,816
model.layers.7.input_layernorm	2,048	688,420,864
model.layers.7.mlp.down_proj	16,777,216	705,198,080
model.layers.7.mlp.gate_proj	16,777,216	721,975,296
model.layers.7.mlp.up_proj	16,777,216	738,752,512
model.layers.7.post_attention_layernorm	2,048	738,754,560
model.layers.7.self_attn.k_proj	1,048,576	739,803,136
model.layers.7.self_attn.o_proj	4,194,304	743,997,440
model.layers.7.self_attn.q_proj	4,194,304	748,191,744
model.layers.7.self_attn.v_proj	1,048,576	749,240,320
model.layers.8.input_layernorm	2,048	749,242,368
model.layers.8.mlp.down_proj	16,777,216	766,019,584
model.layers.8.mlp.gate_proj	16,777,216	782,796,800
model.layers.8.mlp.up_proj	16,777,216	799,574,016
model.layers.8.post_attention_layernorm	2,048	799,576,064
model.layers.8.self_attn.k_proj	1,048,576	800,624,640
model.layers.8.self_attn.o_proj	4,194,304	804,818,944
model.layers.8.self_attn.q_proj	4,194,304	809,013,248
model.layers.8.self_attn.v_proj	1,048,576	810,061,824
model.layers.9.input_layernorm	2,048	810,063,872
model.layers.9.mlp.down_proj	16,777,216	826,841,088
model.layers.9.mlp.gate_proj	16,777,216	843,618,304
model.layers.9.mlp.up_proj	16,777,216	860,395,520
model.layers.9.post_attention_layernorm	2,048	860,397,568
model.layers.9.self_attn.k_proj	1,048,576	861,446,144
model.layers.9.self_attn.o_proj	4,194,304	865,640,448
model.layers.9.self_attn.q_proj	4,194,304	869,834,752
model.layers.9.self_attn.v_proj	1,048,576	870,883,328
model.layers.10.input_layernorm	2,048	870,885,376
model.layers.10.mlp.down_proj	16,777,216	887,662,592
model.layers.10.mlp.gate_proj	16,777,216	904,439,808
model.layers.10.mlp.up_proj	16,777,216	921,217,024
model.layers.10.post_attention_layernorm	2,048	921,219,072
model.layers.10.self_attn.k_proj	1,048,576	922,267,648
model.layers.10.self_attn.o_proj	4,194,304	926,461,952
model.layers.10.self_attn.q_proj	4,194,304	930,656,256
model.layers.10.self_attn.v_proj	1,048,576	931,704,832

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Module	Params	Running total
model.layers.11.input_layernorm	2,048	931,706,880
model.layers.11.mlp.down_proj	16,777,216	948,484,096
model.layers.11.mlp.gate_proj	16,777,216	965,261,312
model.layers.11.mlp.up_proj	16,777,216	982,038,528
model.layers.11.post_attention_layernorm	2,048	982,040,576
model.layers.11.self_attn.k_proj	1,048,576	983,089,152
model.layers.11.self_attn.o_proj	4,194,304	987,283,456
model.layers.11.self_attn.q_proj	4,194,304	991,477,760
model.layers.11.self_attn.v_proj	1,048,576	992,526,336
model.layers.12.input_layernorm	2,048	992,528,384
model.layers.12.mlp.down_proj	16,777,216	1,009,305,600
model.layers.12.mlp.gate_proj	16,777,216	1,026,082,816
model.layers.12.mlp.up_proj	16,777,216	1,042,860,032
model.layers.12.post_attention_layernorm	2,048	1,042,862,080
model.layers.12.self_attn.k_proj	1,048,576	1,043,910,656
model.layers.12.self_attn.o_proj	4,194,304	1,048,104,960
model.layers.12.self_attn.q_proj	4,194,304	1,052,299,264
model.layers.12.self_attn.v_proj	1,048,576	1,053,347,840
model.layers.13.input_layernorm	2,048	1,053,349,888
model.layers.13.mlp.down_proj	16,777,216	1,070,127,104
model.layers.13.mlp.gate_proj	16,777,216	1,086,904,320
model.layers.13.mlp.up_proj	16,777,216	1,103,681,536
model.layers.13.post_attention_layernorm	2,048	1,103,683,584
model.layers.13.self_attn.k_proj	1,048,576	1,104,732,160
model.layers.13.self_attn.o_proj	4,194,304	1,108,926,464
model.layers.13.self_attn.q_proj	4,194,304	1,113,120,768
model.layers.13.self_attn.v_proj	1,048,576	1,114,169,344
model.layers.14.input_layernorm	2,048	1,114,171,392
model.layers.14.mlp.down_proj	16,777,216	1,130,948,608
model.layers.14.mlp.gate_proj	16,777,216	1,147,725,824
model.layers.14.mlp.up_proj	16,777,216	1,164,503,040
model.layers.14.post_attention_layernorm	2,048	1,164,505,088
model.layers.14.self_attn.k_proj	1,048,576	1,165,553,664
model.layers.14.self_attn.o_proj	4,194,304	1,169,747,968
model.layers.14.self_attn.q_proj	4,194,304	1,173,942,272
model.layers.14.self_attn.v_proj	1,048,576	1,174,990,848
model.layers.15.input_layernorm	2,048	1,174,992,896
model.layers.15.mlp.down_proj	16,777,216	1,191,770,112
model.layers.15.mlp.gate_proj	16,777,216	1,208,547,328
model.layers.15.mlp.up_proj	16,777,216	1,225,324,544
model.layers.15.post_attention_layernorm	2,048	1,225,326,592
model.layers.15.self_attn.k_proj	1,048,576	1,226,375,168
model.layers.15.self_attn.o_proj	4,194,304	1,230,569,472
model.layers.15.self_attn.q_proj	4,194,304	1,234,763,776
model.layers.15.self_attn.v_proj	1,048,576	1,235,812,352
model.norm	2,048	1,235,814,400